

Immigration Profile Evaluation Summary

Candidate Name: Vivek Kumar Mishra

Current Title: Senior Software Engineer at JPMorgan Chase & Co.

Petitions Sought: EB-1A (Extraordinary Ability) & EB-2 NIW (National Interest Waiver)

Note: This document provides a consolidated, high-level summary of the achievements, evidence, and criteria claims documented in the petitioner's LaTeX petition drafts. It is designed for attorney review to evaluate the strength of the petitioner's profile.

1. Education & Credentials

- **Master of Science in Computer Science (2021)**
California State University, Long Beach (CSULB) - R2 Doctoral University
- **Bachelor of Technology in Information Technology (2014)**
West Bengal University of Technology, India
- **Certifications:** AWS Certified Cloud Practitioner

2. Proposed Endeavor (for EB-2 NIW)

Endeavor: Applying AI-driven intelligent systems to enhance the resilience and optimization of U.S. critical infrastructure.

- **Financial Infrastructure:** Architecting enterprise-scale, high-availability distributed systems at a Systemically Important Financial Institution (SIFI) protecting 80+ million Americans.
- **Transportation Infrastructure:** Conducting U.S. DOT-funded research on intelligent transportation systems, contributing to the federal Justice40 Initiative.
- **Policy Alignment:** Strongly aligns with U.S. Executive Order 14110 on safe, secure, and trustworthy AI deployment.

3. EB-1A Criteria Evidence

Criterion A: Performance in a Leading or Critical Role

- **JPMorgan Chase & Co. (Senior Software Engineer):**
 - Served a critical role in launching “The Shops at Chase”, a massive e-commerce integration for Chase cardmembers.
 - Architected backend systems processing 5,012,443 monthly requests and 22,623 orders with 99.9% reliability.
 - Work generated over ~\$100,000 in immediate cost savings and a 43% response-time improvement.

- Appreciated globally by VP-level leadership and enterprise architects.
- **California State University, Long Beach (Student Research Assistant):**
 - Hand-selected by the Civil Engineering Department for his specialized Python and AI expertise to bridge infrastructure applications on U.S. DOT-funded federal research projects.

Criterion B: Original Scientific/Scholarly Contributions of Major Significance

- **Technical Framework Authorship:** Authored the acclaimed technical book, *“Design Patterns for AI Agents: A Structured Approach to Building Robust and Intelligent Systems”* (ISBN: 979-8268912371). Formally incorporated into the academic curriculum at **Anand International College of Engineering** (Library Collection) and **Supaul College of Engineering** (B.Tech CSE AI Curriculum).
- **Research Adopted by State Agencies:** Methodologies developed in his transportation equity research have been subsequently utilized/adopted by Florida DOT-funded research.
- **Federal Indexing:** His research was recognized as a permanent contribution to the national transit knowledge base and is officially indexed in the **U.S. National Transportation Library** (ROSA P Records dot/61210 & dot/59030).

Criterion C: Authorship of Scholarly Articles

Impact Metrics: 9 Peer-Reviewed Publications & 37 Citations

- *“Framework for Rail Transport Inequality Assessment...”* (15 citations).
- *“Evaluating Financing Mechanisms and Economic Benefits to Fund Grade Separation Projects”* (12 citations).
- *“AI-Driven Optimization of Backend and Cloud Infrastructure in Large-Scale Financial Systems”* (IJARSCT Journal, Impact Factor: 8.2).
- *“Integrating Solar Photovoltaic Systems into the Grid: An Overview of AI Application”* (MDPI Sustainability Journal, Impact Factor: 3.9).

Criterion D: Participation as a Judge of the Work of Others

- **Premier AI/ML Conferences:**
 - **ICLR 2025** (CORE A*): Conducted review for the FM-Wild Workshop.
 - **NeurIPS 2025** (CORE A*): Assigned multiple papers for the Position Paper Track (e.g., AI automation, AI climate impact).
 - **Other Conferences:** GSP 2025, ACDSA 2026, ICECET 2026 (Awarded Official Certificate of Reviewing).
- **IEEE Panel Service:** Served on the **IEEE Senior Member Application Review Panel** (June 2025) to judge elevation packages of senior engineering professionals globally.

- **Academic Journals:** Reviewer for Virtus InterPress (*Risk Governance and Control, Corporate & Business Strategy Review*) and *Informatica*.
- **Innovation Competitions/Hackathons:** Certified Judge for **PennApps** (University of Pennsylvania), DragonHacks 2024, and Claro 2024.

Criterion E: Membership in Associations Requiring Outstanding Achievements

- **IEEE Senior Member:** Elevated to Senior Member grade (held by only the top 8% of IEEE’s 460,000 global members), necessitating extensive professional practice and 5 years of significant performance.

Criterion F: Published Material About the Beneficiary’s Work

- **High-Traffic Tech Media Profiles & Interviews:** **TechBullion** (230K visitors/mo) exclusive interview, **VoyageLA** feature “Rising Stars: Meet Vivek Mishra”.
- **Expert Feature / Quotes:** Quoted extensively as an IEEE Senior Member expert in articles by **TechTarget** (1.4M+ daily readers), **Dice.com** (2.4M visitors/mo), and **VKTR.com** (7M+ readers).
- **Institutional / Professional Spotlights:** Official Profile on **IEEE Transmitter** as an “IEEE Impact Creator”, **IEEE.tv** Speaker Profile, and explicit recognition posts via **IEEE’s Official X/Twitter** account.
- **Mainstream Regional Newspapers:** Profiled/covered in **Dainik Bhaskar** (India’s #1 Hindi newspaper, 63M readers) and **Rajasthan Patrika** (25M readers).

4. Expert Testimony (Letters of Recommendation)

The applicant secured letters of support from authoritative industry and academic experts verifying his impact:

1. **Corey Graham:** Director of Software Engineering, JPMorgan Chase & Co. (23 years experience).
2. **Sumit Bhatnagar:** VP of Engineering, JPMorgan Chase & Co. (IEEE Senior Member, IETE Fellow).
3. **Dr. Maxim A. Dulebenets:** Associate Professor, FAMU-FSU College of Engineering.
4. **Dr. Shailesh Chandra:** Associate Professor, CSULB.
5. **Dr. Zhenyu Wang:** Senior Research Associate, University of South Florida.
6. **Sinchan Banerjee:** Director I - Enterprise Architecture, UST.
7. **Rahul Vishwakarma:** Research Engineer at WorkOnward / Techstars (holds 60+ U.S. patents).